POWER ELECTRONICS AND DRIVES LAB

(PCEEL408)

Cycle 2

5. **Effect of source inductance in single-phase controlled rectifier feeding inductive load**

   Aim: **Aim:** To set up a single-phase full controlled rectifier with source inductance, for highly inductive loads, observe relevant waveforms and calculate the source power factor, line current and the average voltage lost due to the effect of source inductance. Simulate the circuit using any suitable simulation software and find the input power factor and THD. Compare the results.

6. **a. High frequency diode – Measurement of power loss and reverse recovery time**

   Aim: To measure the power losses & reverse recovery time of a high frequency diode compare with theoretical estimate and with other diodes

   b. Switching characteristics of MOSFET

   Aim: To study and understand the switching characteristics of a power MOSFET, measure rise time, fall time and compute switching losses at the switch during turn on and turn off. Estimate the ON state (conduction) loss at 75° C and switching loss of the given MOSFET at 1kHZ switching frequency.

7. **a. Isolated Gate Driver Circuit for low side MOSFET /IGBT using TLP 250/TLP350/PC817**

   Aim: To setup a gate drive for the MOSFET/IGBT for a certain switching frequency with galvanic isolation using suitable method for ground-referenced configurations, and to observe relevant waveforms and analyse the limitations. To simulate the gate driver circuit and compare the results.

   b. Gate Driver circuit for high side MOSFET/ IGBT with Bootstrap Technique and totem pole/push pull configuration: To setup a gate driver circuit for the MOSFET/IGBT for floating reference configuration. Explore bootstrapping technique and to observe relevant waveforms. Add a push pull configuration compare the switching speed. To simulate the gate driver circuit and compare the results.

8. **a. Rectifier fed PMDC/SEDC Motor Drive:** Set up a 1-phase rectifier for motor speed control; observe continuous conduction waveforms. (Demo)
   b. PCB design and soldering practice (Mandatory): PCB design of one experimental circuit setup using a standard software (KiCAD, Altium, etc.) and soldering of the circuit using through-hole or SMD components. (1. Fully controlled rectifier with variable gate trigger circuit. 2. Buck converter with gate driver circuit and variable pulse generator. 3. Boost converter with gate driver circuit and variable pulse generator. 4. Buck-boost converter with gate driver circuit and variable pulse generator. 5. Inverter with gate driver circuit and variable pulse generator)